



“ ‘OUR PYGMIE’ IS NO MAN, NOR YET THE COMMON APE; BUT A SORT OF ANIMAL THAT BELONGED BETWEEN MAN AND THE APES. ”

Edward Tyson, British physician, in *Orang-Outang, sive Homo Sylvestris* (Orang-Outang, or Man of the Woods), 1699

STEAM POWER

When water is heated to boiling point it creates gaseous steam; if this steam is then trapped in a sealed container and cooled, it condenses back into water. As the quantity of gas drops, so does its pressure, creating a partial vacuum. Force is

then generated when the atmosphere is let in to fill the void. The idea of harnessing this force in an “atmosphere engine” originated in the 1690s and would be fully realized in the steam engines of the next century.

autopsies in an effort to understand the causes of mental illness. But he dissected animals too, so becoming the **father of comparative anatomy**. In 1699, he published his **study of the chimpanzee** (which he called an “orang-outang”), concluding that it had more in common with humans than it did with monkeys.

That year, an English inventor **Thomas Savery** (1650–1715) demonstrated his latest creation to the Royal Society: “an engine to raise water by fire.” Patented the year before, it exploited the recently discovered power of gas pressure, which could generate considerable force when gas rushed in to fill a vacuum.

Savery’s steam pump consisted of a boiler to produce steam that was directed into a vessel below a cold-water shower. This created a vacuum in the vessel as the steam condensed, which sucked up water from below. The

sequence was controlled by a system of taps. Savery claimed his pump could be used to pull water up from mines, but it had a working height limit of about 25 ft (7.5 m). It was also vulnerable to explosion.

Again in 1699, Welsh naturalist **Edward Lhyud** (1660–1709) published a **catalog of fossils**. This included one of the earliest unambiguous specimens—a tooth—later identified as that of a dinosaur. Lhyud had fanciful notions about his specimens, suggesting that fossils grew in rocks from vaporous spawn that came from the sea.

French physicist Guillaume Amontons (1663–1705) was an **accomplished instrument**

Savery’s steam pump
Having understood the principles of atmospheric force, Savery created a steam generator to pump water vertically.

maker and perfected thermometers and barometers for measuring temperature and pressure. He was also the first experimenter to discuss the **idea of an absolute zero for temperature**. In 1699, Amontons turned to mechanics, describing how friction force depended upon load. **Amontons’ friction law** had a prestigious history, being based on experiments first performed by Leonardo da Vinci.



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1699 Guillaume Amontons describes his friction law